Open-Source Development

ITIC2262

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3. Working with the Open-source products

* **Identifying Open-Source Products:**
  - Operating Systems
  - Office Packages
  - Web Browsers
  - Databases
  - Multimedia
  - Development Tools

Intended Learning Outcomes

- By the end of this lecture, students will be able to Working with the Open-source products and Analyze features of the open-source products:
  - Define open-source software products.
  - Identify popular open-source operating systems.
  - Recognize commonly used open-source office applications.
  - Identify open-source web browsers and web servers.
  - Explore other widely used open-source software products.
  - Compare open-source and proprietary software.

What are Open-Source Products

* **Definition**

Open-source products are software applications whose source code is publicly available, allowing users to:

- Use freely
- Study the code
- Modify the software
- Share and distribute copies
- Contribute improvements

* **Key Characteristics**
  - Free to use
  - Community-driven
  - Transparent
  - Customizable
  - Regularly updated

Categories of Open-Source Products

- Open-source software is available in many categories:
  - Operating Systems
  - Office Suites
  - Web Browsers
  - Web Servers
  - Databases
  - Programming Tools
  - Multimedia Applications
  - Graphics & Design Software
  - Content Management Systems (CMS)
  - Learning Management Systems (LMS)

Popular Open Source Operating Systems

* **Linux:**
  The most famous option; it runs servers, supercomputers, and personal computers through versions like Ubuntu or Debian.

* **FreeBSD:**
  A strong, stable system based on Unix, great for networking and high performance.

* **Android:**
  A mobile operating system built on top of the Linux core.

* **Haiku:**
  A small, fast system made for personal use, inspired by older software

Open-Source Operating Systems

* **What is an Operating System?**

An Operating System (OS) manages computer hardware and software resources while providing services for applications.

* **Popular Open-Source Operating Systems**

Operating System	Primary Use
Linux	Servers & Desktops
Ubuntu	Desktop & Education
Debian	Stable Server OS
Fedora	Development & Innovation
Linux Mint	Desktop Computing
Rocky Linux	Enterprise Servers

Features

- Secure
- Stable
- Customizable
- Multi-user support
- Free updates

Open-Source Office Packages

An office package provides tools for creating and managing documents, spreadsheets, and presentations.

* **Popular Office Suites**

Software	Main Features
LibreOffice	Writer, Calc, Impress
Apache OpenOffice	Word Processing, Spreadsheet, Presentation
ONLYOFFICE Community Edition	Office Collaboration
Calligra Suite	Productivity Applications

* **Applications**
  - Documents
  - Spreadsheets
  - Presentations
  - Databases
  - Drawing

Open-Source Web Browsers

A web browser allows users to access websites and web applications.

* **Popular Open-Source Browsers**

Browser	Features
Mozilla Firefox	Privacy & Security
Chromium	Fast & Lightweight
Brave (Open-source core)	Privacy-focused
Tor Browser	Anonymous Browsing

Features

- Tab browsing
- Extensions
- Password manager
- Private browsing
- Developer tools

Open-Source Web Servers

A web server hosts websites and web applications.

* **Popular Web Servers**

Server	Usage
Apache HTTP Server	Most popular web server
Nginx	High-performance server
Lighttpd	Lightweight web server
Caddy	Automatic HTTPS support

* **Common Uses**
  - Website hosting
  - APIs
  - Cloud services
  - Enterprise application

Open-Source Database Systems

Databases store and manage data efficiently.

* **Popular Databases**

Database	Type
MySQL	Relational Database
PostgreSQL	Advanced Relational Database
MariaDB	MySQL Alternative
SQLite	Embedded Database
MongoDB Community Edition	NoSQL Database

* **Applications**
  - Banking
  - Education
  - Healthcare
  - E-commerce

Open-Source Programming Tools

* **Development Environments**
  - Eclipse IDE
  - Apache NetBeans
  - Visual Studio Code (open-source core)
  - Code::Blocks
* **Version Control**
  - Git
* **Repository Hosting**
  - GitHub
  - GitLab
  - Codeberg

Open-Source Multimedia Software

* **Media Players**
  - VLC Media Player
  - MPV
* **Audio Editing**
  - Audacity
  - Ardour
* **Video Editing**
  - Kdenlive
  - OpenShot
* **Streaming**
  - OBS Studio

Open-Source Graphics & Design Software

Software	Purpose
GIMP	Image Editing
Inkscape	Vector Graphics
Blender	3D Modeling & Animation
Krita	Digital Painting
Scribus	Desktop Publishing

Open-Source Content Management Systems (CMS)

A CMS helps users build and manage websites without extensive programming.

* **Popular CMS**
  - WordPress
  - Drupal
  - Joomla

* **Applications**
  - Business websites
  - Blogs
  - News portals
  - Educational websites

Open-Source Learning Management Systems (LMS)

* **Popular LMS Platforms**
  - Moodle
  - Open edX
  - Canvas LMS (Open Source)

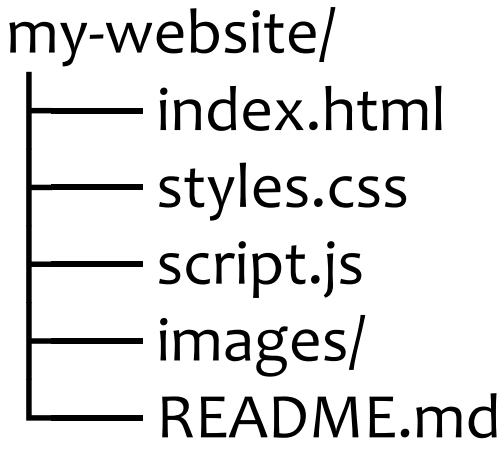
* **Used For**
  - Online learning
  - Course management
  - Assignments
  - Quizzes
  - Student assessment

A GitHub repository

- A **GitHub repository** (often called a **repo**) is a storage space on GitHub where you keep a project's files, code, and complete change history.

- Think of it like a project folder with superpowers:
  - **Stores files** such as source code, images, documentation, and configuration files.
  - **Tracks changes** using Git, so you can see what changed, when, and by whom.
  - **Supports collaboration**, allowing multiple people to work on the same project without overwriting each other's work.
  - **Keeps history**, so you can restore previous versions if something goes wrong.

* For example, a repository for a website might look like this:



* A typical GitHub repository also includes features like:
  - **README.md –** Explains what the project is and how to use it.
  - **Issues –** Track bugs, feature requests, and tasks.
  - **Pull requests –** Review and discuss proposed changes before merging them.
  - **Actions –** Automate testing, building, and deployment.
  - **Wiki –** Store project documentation.

Git vs. GitHub

* It's helpful to distinguish the two:
  - **Git** is the version control software that tracks changes on your computer.
  - **GitHub** is an online platform that hosts Git repositories and adds collaboration features.
* Example
  - Suppose you're writing a Python calculator:
    - ✓ Create a Git repository on your computer.
    - ✓ Add your files (calculator.py, README.md, etc.).
    - ✓ Push the repository to GitHub.
    - ✓ Continue making changes over time. Git records each version, and GitHub lets others view, contribute to, or review your work.

Open Source vs Proprietary Software

Feature	Open Source	Proprietary
Source Code	Available	Not Available
Cost	Usually Free	Paid License
Customization	High	Limited
Community Support	Available	Vendor Support
Licensing	Open Licenses	Commercial Licenses

Advantages of Open-Source Products

* **Technical Benefits**
  - Reliable
  - Secure
  - Flexible
  - Transparent

* **Economic Benefits**
  - No licensing fees
  - Lower operational costs

* **Educational Benefits**
  - Learn from source code
  - Community support
  - Encourages innovation

Challenges of Open-Source Products

- Limited vendor support
- Learning curve
- Compatibility issues
- Documentation quality varies
- Maintenance responsibility
- Some specialized commercial features may be absent

Questions

1. What is open-source software?
2. Name three open-source operating systems.
3. Which open-source office suite have you used?
4. What are two open-source web browsers?
5. Why do organizations choose open-source software?
6. Compare open-source software with proprietary software.
7. Which open-source product would you recommend for students and why?

Practical Lab Activity

Install and explore any of the following open-source applications:

- **Ubuntu/Centos** (or use a live USB/virtual machine)
- **LibreOffice**
- **Mozilla Firefox**
- **VLC Media Player**
- **GIMP**
- **Git**
- **Visual Studio Code** (or **Apache NetBeans**)
- **Moodle** (demo site, if available)

Students should complete simple tasks such as **creating a document, browsing the web, editing an image, playing a media file, and exploring a Git repository** to gain hands-on experience with open-source products.

Practical

- Booting Centos Linux
- Creating a document in LibreOffice Writer
- Browsing with Mozilla Firefox
- Playing media using VLC
- Editing an image using GIMP
- Viewing a GitHub repository – (https://github.com/login?return_to=https%3A%2F%2Fgithub.com%2Fnew)

THANK YOU